

MEKAL BUTT

COMPUTER SCIENCE & SYSTEMS ENGINEERING

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL OBJECTIVE

A dedicated 5th-semester Computer Science student at the University of Management and Technology, working towards a specialization in Cybersecurity and Computer Networks. Backed by 4+ years of foundational experience in C++ logic building, alongside specialized expertise in modern AI workflows and autonomous automation using Python and Cursor. Demonstrated hands-on capability through complex academic projects spanning database design, data structures, and enterprise network deployment. Actively seeking a remote part-time role or technical internship to apply strong technical adaptability, rapid learning, and a commitment to systems defense in solving real-world challenges.

EDUCATION

University of Management & Technology <i>Bachelor of Science in Computer Science (BSCS) • 5th Semester • Lahore, Pakistan</i>	2024 – 2028
KIPS College <i>Intermediate Computer Science (ICS) • Marks: 803/1100</i>	2022 – 2023

TECHNICAL COMPETENCIES

Programming: C++, C, Object-Oriented Programming (OOP)

AI & Automation: Agentic AI & Autonomous Agents, AI Pair Programming (Python), Prompt Engineering, API Integration

Networking: Cisco Packet Tracer, Network Topology, Routing Protocols, Security & Basic Protocols, Server/PC Configurations, Subnetting

Database: SQL, DBMS, Relational Design, SaaS

Core Concepts: Data Structures & Algorithms (DSA), Software Engineering

Security: Vulnerability Assessment (Beginner)

PROJECTS & PRACTICAL EXPERIENCE

Agentic AI: YT Autonomous Media Processing Engine <i>Python Automation & API Integration</i>	July – Aug 2026
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[View Architecture on GitHub](#) [View on Portfolio](#)

- Engineered a fully autonomous, state-aware media processing pipeline using Python, applying strict memory management and Just-In-Time (JIT) execution principles.
- Integrated FFmpeg, Gemini Multimodal AI, and the YouTube Data API to dynamically compile, render, and schedule 9:16 vertical video content for a gaming YouTube channel.
- Designed a hardware-level OS interrupt trigger and a dynamic queue allocator to ensure self-sustaining, API-quota-safe execution over infinite lifecycles.

Integrated Project Management Database (DBMS) <i>Relational Architecture & SQL Development</i>	March 2026
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[View ERD \(Database Schema\) on Portfolio](#)

- Designed and implemented a normalized relational database schema for an Integrated Project Management System.
- Developed SQL queries, joins, constraints, and data retrieval operations.
- Applied DBMS concepts alongside C++, OOP, and DSA to ensure data integrity and optimize query performance.

SERVICES & OFFERINGS

- Database Design & SQL Development** - Schema design, normalization, and query writing, with or without AI-assisted tools.

- **C++ / OOP Software Development** - Building, testing, and debugging object-oriented applications.
- **Data Structures & Algorithm Problem-Solving** - Delivering efficient solutions with time and space complexity analysis.
- **Network Design & Configuration** - Topology planning and subnetting using Cisco Packet Tracer.
- **AI-Assisted Programming & Debugging (Python)** - Utilizing Agentic AI and pair-programming tools to accelerate development cycles.
- **Technical Content Writing** - Producing clear documentation, reports, and explainer articles on Computer Science topics.

CERTIFICATIONS

- **Cisco Networking Academy:** Specialized coursework encompassing foundational and intermediate networking architectures, security protocols, and their respective tools and environments.
- **Intro to Networks by Cisco** (Under Progress)